OTH REGENSBURG

Praktikum Regelungstechnik (RTP)[TRRT, RSSCRIPT]

Versuch 100:

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Ausarbeitung als Beispiel

Gruppe 38
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`Dokument-Version: 02`

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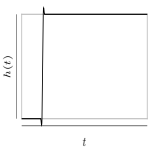
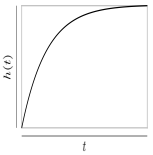
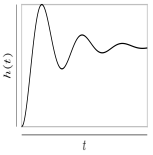
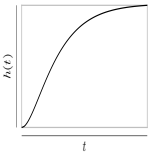
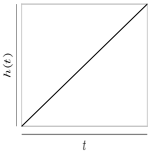
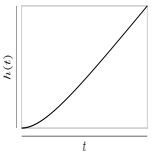
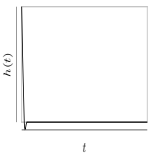
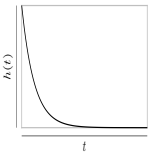
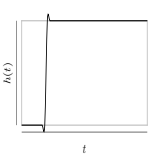
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A. Appendix A **18**

1 Preperations

Zur Vorbereitung auf diesen Versuch wurden die in ELO bereitgestellten Dateien heruntergeladen und gespeichert [**RTP˙Versuche**]. Zudem wurden die jeweiligen Sprungantworten verschiedener Systemverhalten recherchiert:

1.1. Different basic system behaviors

Systemverhalten	Sprungantwort	Visualization
P-Glied (K_p)	$G(s) = K_p$ (1.1)	
PT_1-Glied (K_p, T_1)	$G(s) = \frac{K_p}{T_1 s + 1}$ (1.2)	
PT_2-Glied (K_p, ω_0, D)	$G(s) = \frac{K_p}{\frac{1}{\omega_0^2} s^2 + \frac{2D}{\omega_0} s + 1}$ (1.3)	
PT_2-Glied für $D \geq 1$ (K_p, T_1, T_2)	$G(s) = \frac{K_p}{(T_1 s + 1)(T_2 s + 1)}$ (1.4)	
I-Glied (K_I)	$G(s) = \frac{K_I}{s}$ (1.5)	
IT_1-Glied (K_I, T_1)	$G(s) = \frac{K_I}{s(T_1 s + 1)}$ (1.6)	
D-Glied (K_D)	$G(s) = K_D\, s$ (1.7)	
DT_1-Glied (K_D, T_1)	$G(s) = \frac{K_D\, s}{T_1 s + 1}$ (1.8)	
T_t-Glied (K_p, T_t)	$G(s) = K_p\, e^{-T_t s}$ (1.9)	

1.1.1. P-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.2. PT_1-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.3. PT_2-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.4. I-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.5. IT_1-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.6. D-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.7. DT_1-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.1.8. T_t-Glied

Explanation

Parameters needed for characterization

Step and Impulse response

1.2. Identified example systems

1.2.1. System $N = 0$

Step and Impulse response

Differnetal Equiation

Transfer Function

1.2.2. System $N = 1$

Step and Impulse response

Differnetal Equiation

Transfer Function

1.2.3. System $N = 1$

Step and Impulse response

Differnetal Equiation

Transfer Function

2 Experiment T2 — System identification

2.1. System $N = 0$

2.1.1. Experimentaly Identified system parameters

2.1.2. Documentation of the results

Essential Commands

Differential Equation

Transfer Function

2.1.3. Realworld Example

description of the respective input and output variables.

2.1.4. Comparison of given System with the response of the identified system

Deviations of the System

3 Conclusion

A Appendix A